Progressive Education Society's
MODERN COLLEGE OF ENGINEERING, Pune -05.
(An Autonomous Institute Affiliated to Savitribai Phule Pune University)
MCA Department

PRACTICAL SUBMISSION RECORD- A.Y. 2025-26

Class: SYMCA **Div.: B** **Semester: III**	**Course Code: MCA01604** **Course Name: Data Science Laboratory**	**Batch: S2**
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CO No: CO605.1	**Assignment No: 5**	

Title: Use a dataset with features like square footage vs. house price. Implement a simple linear regression model to predict house prices. Plot the regression line and calculate MSE and R^2 score.

Code:-

```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score

np.random.seed(42)
square_footage = np.random.randint(1000, 4000, 50)
price = 50000 + square_footage * 150 + np.random.randint(-50000, 50000, 50)

X = square_footage.reshape(-1, 1)
y = price

model = LinearRegression()
model.fit(X, y)
y_pred = model.predict(X)

mse = mean_squared_error(y, y_pred)
r2 = r2_score(y, y_pred)

plt.style.use('seaborn-v0_8-whitegrid')
plt.figure(figsize=(10, 6))
plt.scatter(X, y, color='blue', label='Actual House Prices')
plt.plot(X, y_pred, color='red', linewidth=2, label='Regression Line (Predictions)')
plt.title('House Price vs. Square Footage')
plt.xlabel('Square Footage')
plt.ylabel('House Price ($)')
plt.legend()
plt.grid(True)

print(f"Model Performance:")
print(f"  - Intercept (b₀): ${model.intercept_:.2f}")
print(f"  - Coefficient (b₁): ${model.coef_[0]:.2f} per sq ft")
print(f"  - Mean Squared Error (MSE): {mse:,.2f}")
print(f"  - R-squared (R²) Score: {r2:.4f}")

plt.show()
```

Output:-

```
Model Performance:
  - Intercept (b₀): $36024.05
  - Coefficient (b₁): $157.38 per sq ft
  - Mean Squared Error (MSE): 766,332,280.87
  - R-squared (R²) Score: 0.9591
```